Module:01 (5) 21/04/25

Errors and Approximations

Definitions of approximation errors and examples

Bairstow method:

Problems

1. using Bairstow method obtain the quadratic factor of the following equation

i) $x^4 - 3x^3 + 20x^2 + 44x + 54 = 0 \quad (p, q) = (2, 2)$

Soln: $(p_0, q_0) = (2, 2)$

$$\Delta p_k = -\left[\frac{b_n c_{n-3} - b_{n-1} c_{n-2}}{c_{n-2} c_{n-2} - c_{n-3} c_{n-1} + c_{n-3} b_{n-1}}\right]$$

$$\Delta q_k = -\left[\frac{b_{n-1} c_{n-1} - b_{n-1} b_{n-1} - b_n c_{n-2}}{c_{n-2} c_{n-2} - c_{n-3} c_{n-1} + c_{n-3} b_{n-1}}\right]$$

$n = 4$,

$$\Delta p_k = -\left[\frac{b_4 c_1 - b_3 c_2}{c_2^2 - c_1 c_3 + c_1 b_3}\right] \qquad \Delta q_k = -\left[\frac{b_3 c_3 - b_3^2 - b_4 c_2}{c_2^2 - c_1 c_3 + c_1 b_3}\right]$$

	x^4	x^3	x^2	x^1	x^0
-2	1	-3	20	44	54
			10	-56	4
-2		-2			
			-2	10	-56
-2	1	-5	28	-2	2
	b_0	b_1	b_2	b_3	b_4
-2		-2	14	-80	
			-2	14	-80
	1	-7	40	-68	
	c_0	c_1	c_2	c_3	

(diagonal)

$$K=0 : \Delta p_0 = -\left[\frac{2\times -7 - (-2\times 40)}{40^2 - (-7)(-68) + (-7\times -2)}\right] = -0.058$$

$$\Delta q_0 = -\left[\frac{(-2\times -68) - (-2)^2 - 2\times 40}{40^2 - 7\times 68 + 14}\right] = -0.0457$$

$$P_1 = P_0 + \Delta P_0 = 2 - 0.058 = 1.942$$

$$q_1 = q_0 + \Delta q_0 = 2 - 0.0457 = 1.9543$$

	x^4	x^3	x^2	x^1	x^0
-1.942	1	-3	20	44	54
		-1.942	9.5974	-53.6829	0.0480
-1.9543			-1.9543	9.6582	-54.0229
	b_0	b_1	b_2	b_3	b_4
-1.942	1	-4.9420	27.6431	-0.0247	0.0251
-1.9543		-1.942	13.3687	-75.8497	
			-1.9543	13.4534	-76.3301
	1	-6.8840	39.0575	-62.4210	
	c_0	c_1	c_2	c_3	

$$K=1,\ \Delta P_1 = -\left[\frac{0.0251 \times -6.8840 - (-0.0247 \times 39.0575)}{(39.0575)^2 - (-6.8840 \times -62.4210) + (-6.8840 \times -0.0247)}\right]$$

$$\Delta P_1 = \frac{-0.7919}{10950.9522} = -0.0007$$

$$\Delta q_1 = -\left[\frac{(-0.0247 \times -62.4210) - (-0.0247)^2 - (0.0251 \times 39.0575)}{(39.0575)^2 - (-6.8840 \times -62.4210) + (-6.8840 \times -0.0247)}\right]$$

$$\Delta q_1 = \frac{-0.5608}{10950.9522} = \Delta q_1 = -0.0005$$

$$p_2 = P_1 + \Delta P_1 = 1.942 - 0.0007 \approx 1.9413$$

$$q_2 = q_1 + \Delta q_1 = 1.9543 - 0.0005 = 1.9538$$

2) $x^4 - x^3 + 6x^2 + 5x + 10 = 0$ $P_0 = 1.14$, $a_0 = 1.42$ 22/04/25

$(P_0, a_0) = (1.14, 1.42)$

$n = 4$

soln:-

	x^4	x^3	x^2	x^1	x^0
-1.14	1	-1	6	5	10
-1.42		-1.14	2.4396	-8.0023	-0.0416
			-1.42	3.0388	-9.9678
	1	-2.14	7.0196	0.0365	-0.0094
-1.14	b_0	b_1	b_2	b_3	b_4
-1.42		-1.14	3.7392	-10.6462	6.7854
			-1.42	4.6576	-13.2611
	1	-3.28	9.3388	-5.9521	
	c_0	c_1	c_2	c_3	

$k=0$, $\Delta P_0 = -\left[\dfrac{b_4 c_1 - b_3 c_2}{c_2^2 - c_1 c_3 + c_1 b_3}\right] = -\left[\dfrac{-0.0094 \times -3.28 - 0.0365 \times 9.3388}{(9.3388)^2 - (-3.28 \times -5.9521) + (-3.28 \times 0.0365)}\right]$

$\Delta a_0 = -\left[\dfrac{b_3 c_3 - b_3^2 - b_4 c_2}{c_2^2 - c_1 c_3 + c_1 b_3}\right]$

$= \dfrac{-(-0.3100)}{67.5706}$

$\boxed{\Delta P_0 = 0.0046}$

$\Delta a_0 = -\left[\dfrac{(0.0365 \times -5.9521) - (0.0365)^2 - (-0.0094 \times 9.3388)}{(9.3388)^2 - (-3.28 \times -5.9521) + (-3.28 \times 0.0365)}\right]$

$\Delta a_0 = \dfrac{-(-0.1308)}{67.5706} = \Delta a_0 = +0.0019$

$P_1 = P_0 + \Delta P_0 = 1.14 + 0.0046 = 1.1446$

$a_1 = a_0 + \Delta a_0 = 1.42 + 0.0019 = 1.4219$

x	x^4	x^3	x^2	x^1	x^0	
-1.1446	1	-1	6	5	10	
		-1.1446	2.1547	-8.0541	0.0147	
-1.4219			-1.4181	3.0413	-9.9786	0.0182
-1.1446	1	-2.1446	7.0366	-0.0128	0.0361	
	b_0	b_1	b_2	b_3	b_4	
		-1.1446	3.7648	-10.7401		
-1.4181			-1.4181	4.06644	-13.3069	
	1	-3.2892	9.3833	-6.0885		
	c_0	c_1	c_2	c_3		

$$k=1, \quad \Delta p_1 = -\left[\frac{b_4 c_1 - b_3 c_2}{c_2^2 - c_1 c_3 + c_1 b_3}\right]$$

$$\Delta p_1 = -\left[\frac{(0.0361 \times -3.2892) - (-0.0128 \times 9.3833)}{(9.3833)^2 - (-3.2892 \times -6.0885) + (-3.2892 \times -0.0128)}\right]$$

$$\Delta p_1 = \frac{-0.00014}{68.0621} = 0$$

$$\Delta q_1 = -\left[\frac{b_3 c_3 - b_3^2 - b_4 c_2}{c_2^2 - c_1 c_3 + c_1 b_3}\right]$$

$$\Delta q_1 = -\left[\frac{(-0.0128 \times -6.0885) - (-0.0128)^2 - (0.0361 \times 9.3833)}{(9.3833)^2 - (-3.2892 \times -6.0885) + (-3.2892 \times -0.0128)}\right]$$

$$\Delta q_1 = \frac{+0.2610}{68.0621} = \Delta q_1 = 0.0038$$

$P_2 = P_1 + \Delta P_1 = 1.1446 + 0 = 1.1446$

$q_2 = q_1 + \Delta q_1 = 1.4181 + 0.0038 = 1.4219$

	x^4	x^3	x^2	x^1	x^0
	1	-2	6	5	10
-1.1446		-1.1446	8.4547	-8.0497	0.0003
-1.4219			-1.4219	3.0494	-9.9999
-1.1446	1	-2.1446	7.0328	-6.0003	0.0004
	b_0	b_1	b_2	b_3	b_4
-1.4219		-1.1446	3.7648	-10.7314	
			-1.4219	4.6769	-13.3313
	1	-3.2892	9.3757	-6.0548	
	c_0	c_1	c_2	c_3	

$$k=1,\quad \Delta P_1 = -\left[\frac{0.0004 \times -3.2892 - (-0.0003 \times 9.3757)}{(9.3757)^2 - (-3.2892 \times -6.0548) + (-3.2892 \times -0.0003)}\right]$$

$$\Delta P_1 = \frac{-0.0015}{67.9893} = \Delta P_1 = 0$$

$$\Delta q_1 = -\left[\frac{(-0.0003 \times -6.0548) - (-0.0003)^2 - (0.0004 \times 9.3757)}{(9.3757)^2 - (-3.2892 \times -6.0548) + (-3.2892 \times -0.0003)}\right]$$

$\Delta q_1 = 0$

$P_2 = P_1 + \Delta P_1 = 1.1446$

$q_2 = q_2 + \Delta q_2 = 1.4219$

Graeffe's root-squaring method:-

1. using Graeffe's method find the real roots of the equation

$x^3-6x^2+11x-6=0$

$f(x)=x^3-6x^2+11x-6$

$f(-x)=-x^3-6x^2-11x-6$

$f(x)\cdot f(-x)=-x^6-\cancel{6x^5}-11x^4-\cancel{6x^3}+\cancel{6x^5}+36x^4+\cancel{66x^3}+36x^2$
$-11x^4-\cancel{66x^3}-121x^2-\cancel{66x}+\cancel{6x^3}+36x^2+\cancel{66x}$
$+36$

$f(x)\cdot f(-x)=-x^6+14x^4-49x^2+36$

$-(x^6-14x^4+49x^2-36)$

let $x^2=y$

$g(y)=y^3-14y^2+49y-36$

$\sqrt{\frac{36}{49}}, \sqrt{\frac{49}{14}}, \sqrt{\frac{14}{1}}$

$0.8571, \quad 1.8708, \quad 3.7417$

$g(y)g(-y)=(y^3-14y^2+49y-36)(-y^3-14y^2-49y-36)$

$-y^6-\cancel{14y^5}-49y^4-\cancel{36y^3}+\cancel{14y^5}+196y^4+\cancel{686y^3}+504y^2$
$-49y^4-\cancel{686y^3}-2401y^2-\cancel{1764y}+\cancel{36y^3}+504y^2$
$+\cancel{1764y}+1296$

$g(y)\cdot g(-y) = -y^6 + 98y^4 - 1393y^2 + 1296$

$= -(y^6 - 98y^4 + 1393y^2 - 1296)$

Let $y^2 = z$

$h(z) = z^3 - 98z^2 + 1393z - 1296$

$\left(\frac{1296}{1393}\right)^{1/8}, \left(\frac{1393}{98}\right)^{1/8}, \left(\frac{98}{1}\right)^{1/8}$

$= 0.9821, \quad 1.9417, \quad 3.1463$

Alternative method:

	1	-6	11	-6
square	1	36	121	36
$\times -2$		-22	-72	
1/2	1	14	49	36
		-98	-1008	
1/4	1	98	1393	1296
	1	9604	1940449	1679616
		-2786	-254016	
1/8	1	6818	1686433	1679616
	1	46485124	2844056263000	2821109907000
		-3372866	-22903243780	
1/16	1	43112258	2821153019000	2821109907000

It	x_1	x_2	x_3
$1/2$	$\sqrt{14}$	$\sqrt{49/14} =$	$\sqrt{36/49}$
	3.7417	1.8708	0.8571
$1/4$	$(98)^{1/4}$	$\left(\frac{1393}{98}\right)^{1/4} =$	$\left(\frac{1296}{1393}\right)^{1/4} =$
	3.1463	1.9417	0.9821
$1/8$	$(6818)^{1/8}$	$\left(\frac{1686433}{6818}\right)^{1/8} =$	$\left(\frac{1679616}{1686433}\right)^{1/8}$
	3.0144	1.9914	0.9925
$1/16$	$(43112258)^{1/16}$	$\left(\frac{2821153019000}{43112258}\right)^{1/16}$	$\left(\frac{2821109907000}{2821153019000}\right)$

3.0003 1.9998 1

Roots are 3, 2, 1

⑨ solve $x^6 - 4.8x^4 + 3.3x^2 - 0.05 = 0$

Put $x^2 = t$

put $x^2 = t$

$$t^3 - 4.8t^2 + 3.3t - 0.05 = 0$$

1.9931

	1	-4.8	3.3	-0.05
	1	23.04	10.89	0.0025
		-6.6	-0.4800	
1/2	1	16.44	10.41	0.0025
	1	270.2736	108.3681	0
		-20.82	-0.0822	
1/4	1	249.4536	108.2859	0
	1	62227.0986	11725.8361	0
		-216.5718	0	
1/8	1	62010.5268	11725.8361	0
	1	3845305434	137495232.2	0
		-23451.6722	0	
1/16	1	3845281982	137495232.2	0

$1/2 \Rightarrow$ 4.0546, 0.7957, 0.0155

$1/4 \Rightarrow$ 3.9742, 0.8117, 0

$1/8 \Rightarrow$ 3.9724, 0.8121, 0

$1/16 \Rightarrow$ 3.9724, 0.8121, 0

$x = \pm 1.9931, \pm 0.9012, 0$

The roots are $1.9931, -1.9931, +0.9012, \cancel{4}\ -0.9012, 0, 0$